Symmetry Is Not Localised in Parameter Space

NeurReps 2026 — paper outline. Workshop format (~9pp incl. refs), can be cut down to a 4 page extended abstract.

Thesis

> The apparent localisation of a learned symmetry to a few parameters is an artifact of the attribution construction. Measured against matched nulls, the symmetry-generator direction is among the **least** localised directions in a converged network's tangent space — and it becomes less localised, not more, as the network becomes equivariant.

Everything below serves that sentence. The negative half is fully established (theorem + measurement, two models). The positive half — *delocalisation over training* — is the payoff and is what makes this a result rather than a critique.

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1. Introduction (0.75pp)

- The question: for a network that has learned a physical symmetry, can we say *which parameters* realise it? Recent work (our own SCML draft included) answers by projecting the generator onto the functional tangent space and reading off minimum-norm coefficients `c = J⁺g`, observing that `|c_i|` is sharply concentrated.
- We show that concentration is not evidence about symmetry: it is reproduced exactly by a random target and by the Jacobian's conditioning with no target at all. The underlying reason is structural, not statistical.
- Contributions:
    1. An exact factorisation of the attribution into a symmetry-dependent and a **target-independent** conditioning factor, and the observation that the latter dominates after convergence ($R^2 = 0.91$).
    2. A gauge analysis: `Jc = g` is underdetermined by ~4500 dimensions, and the two most natural norm choices return **disjoint** parameter sets.
    3. Causal tests with sensitivity-matched controls, showing the published score has *exactly zero* causal effect on equivariance.
    4. The positive result: symmetry-specific concentration is real in an untrained network and vanishes as the network becomes equivariant — symmetry control is absorbed into task control.
    5. Concrete recommendations for what to report instead, all of which we show to be invariant.

2. Setup (0.75pp)

- `f_θ`, sensitivities `S_i`, tangent space `T_θ = Im(J_θ)`, truncated SVD.
- Generator `X`, intrinsic direction `g = Xf_θ`; group average `Π`, defect `δ = (I−Π)f_θ`.
- **Prop. 0** (from Torben's critique): `⟨δ, g⟩ = 0`. `g` is tangent to the group orbit at constant distance from `ker X` — it measures orbit phase, not distance from equivariance. Attribution of `g` therefore cannot license claims about symmetry *breaking*; that requires `δ`.
- Benchmarks: Mexican-hat (SO(2)) and two-body (rotation only — `Π` needs a compact group, so translation is out of scope by construction). ASRNN and MLP. Two models at matched convergence (loss 6.4–6.7e-5, equivariance residual 4.2e-3 and 9.1e-3) — squarely the regime the claim is about.

3. The attribution is a gauge choice (2pp — theory core)

3.1 Underdetermination. rank(J) = 22 against 4514 parameters (2221 live). Solution set is 4492-dimensional; every point reproduces the identical `P_T g`. Refining the probe grid 96 → 1792 rows leaves rank at 22 — intrinsic, not undersampling.

3.2 Prop. 1 (conditioning–alignment factorisation). With column normalisation, `c_i = a_i / ‖S_i‖²` where `a_i = ⟨V_{:,i}, u⟩` is the only target-dependent factor. Verified 6×10^{1}. **Corollary**: the conditioning factor cancels *exactly* in a ratio of attributions against two targets, giving a provably conditioning-free score (verified 4×10^{1}).

3.3 Prop. 2–4 (gauge and covariance). Under `θ_i ↦ λ_iθ_i` the true coefficients must transform as `c ↦ Λc`. Three independent sources break this: the minimised norm (`‖c‖₂` is not covariant); the *relative* SVD truncation (rank 22 vs 19 under the same reparametrisation, and covariance returns only at cutoff 10^{12}, where truncation no longer does its job); and — an implementation point worth fixing — a dead-column floor defined relative to the largest column.

3.4 Prop. 5 (no neutral gauge). Under `‖D^s c‖₂` the explicit factor is `d_i^{-2s}`, so every choice takes a side: `s=0` (the published `J⁺g`) favours the *most* sensitive parameters (=+0.92), `s=1` the *least* (=−0.88), `s=½` is neutral (=+0.04).

3.5 Prop. 6 (permutation). Hidden-unit permutation leaves `f_θ` unchanged, so parameter indices are not well-posed targets for localisation. The hidden unit is the finest permutation-equivariant granularity.

3.6 Prop. 7 (the contrast). `E_i` is built from column norms of two Jacobians with no optimisation and no truncation — exactly invariant (2.8×10^{1}) under the same reparametrisation that moves `c_i` by ~1.0.

4. What the measurements show (2.5pp — experiments)

4.1 The concentration is not about symmetry. Top 10% of parameters carry 99.9% of `|c_i|` for the true generator — 99.9% for a matched-construction null, 100.0% for `1/‖S_i‖²` alone. R^2(log|c_i| ~ log‖S_i‖) = 0.906, and this *worsens with convergence* (0.15 on a lightly-trained model → 0.91 at convergence, because training spreads `‖S_i‖` over eight decades).

4.2 Gauge disagreement, measured. Mean cross-gauge top-20 Jaccard 0.140; the published `J⁺g` and the column-normalised default share **zero** parameters. The proposed scale-invariant fix `ã_i` removes the conditioning confound exactly (slope −0.000) but not the gauge problem (0.187).

4.3 Causal tests. Order parameter `D = ‖δ‖²/‖f‖²`, differentiable, with `∇_θD` as exact per-parameter causal ground truth, and a **sensitivity-matched control** (nearest-neighbour in log‖S_i‖) that validates itself — `‖S_i‖`, which carries no symmetry information, scores zero excess. - Conditioning alone predicts causal influence at = +0.934. - The published score's top-20, perturbed, gives **Δlog D = 0.0000** — exactly no effect, because that gauge selects near-dead parameters. - Alignment `|⟨S_i, g⟩|` does show a positive excess over the matched control, replicated across both models. - Disjoint blocks by alignment rank each show positive excess at comparable magnitude, and the top block is not privileged — the controlling set is causally real but **not unique**, as ~100-fold substitutability (2221 live parameters, rank 22) predicts.

4.3b Identifiability under resampling. Bootstrap 80% of probe rows, 40 draws, with two reference points: `‖S_i‖` (target-blind, trivially stable — the "easy" level) and a **null-vs-null calibration** `r′` with the same estimator and denominator noise as the conditioning-free score but no symmetry signal — the noise floor.

score	top-20 Jaccard, parameters	top-19, hidden units
`‖S_i‖` (conditioning reference)	0.776	0.935
`|c_i|` (the published score)	**0.748**	0.881
`r_i` (conditioning-free)	0.072	**0.514**
`r′_i` (null-vs-null control)	0.063	0.397

`|c_i|` looks robust and lands within noise of the conditioning reference it is largely made of; the conditioning-free part sits at its own noise floor. Only at unit granularity does the symmetry-specific score clear its control. Together with §3.1 (grid refinement) and §4.2 (gauge), this is the third independent leg under "the parameter-level identity is not a property of the network".

4.4 Sufficiency vs. influence — and both point the same way. These are distinct questions and we test both. - *Influence* is diffuse: n_eff(D) 365 of 2221, versus 837 for the task loss — barely more concentrated than the network's general behaviour. - *Sufficiency*, via gauge-free greedy subset selection against matched nulls: the true generator needs **more** parameters than a non-symmetry target (19 vs 3.2 for 10% error; z = −16). In a near-equivariant network `g` is a small unstructured residual spread across the tangent space, while a generic target loads onto the dominant singular directions. - Both routes agree, which is the thesis.

4.5 The positive result: delocalisation over training. Matched null `E(M) = □ f(Mx) − Mf(x)□ ²/□ f□ ²` for random non-orthogonal `M` — same functional as the true rotation, defined without a group average.

	equivariance error	concentration z vs null	(symm, task)
untrained	1.50	**+3.00**	−0.06
trained	8.2×10	**+0.24**	+0.30

At initialisation symmetry control is concentrated in a symmetry-specific way and is decoupled from task control. After training to equivariance it is indistinguishable from the null and substantially coextensive with the task. Note the *function* is emphatically symmetric (E under the true rotation is 8×10 against 8×10 ² for random `M`): it is the localisation of control, not the symmetry, that dissolves. **Interpretation:** training on a symmetric system does not carve out a symmetry subnetwork — it absorbs symmetry into general task competence.

5. What to report instead (0.75pp)

- **`E_i`** — exactly invariant; the published Fig. 1b stands without qualification where Fig. 1a does not.
- **Function-space quantities**: `P_T g`, the representation residual `□ P_{T□ }g□ /□ g□` (currently unreported, and it *degrades* to 0.12–0.24 in the low-equivariance regime), and `□ g□ , □ δ□` as order parameters.
- **rank(J)** as the well-posed concentration statement — gauge-, reparametrisation-, permutation-, and grid-invariant.
- **Hidden-unit granularity**, via set overlap rather than correlation. (Module- level *correlation* is saturated — a null-vs-null control scores 0.995 — and must not be over-claimed. The genuine signal is unit-level Jaccard: 0.514 vs null 0.397, and 0.65 vs 0.09 under grid refinement.)
- **A matched null in every figure** — the difference between a claim about symmetry and a claim about conditioning.

6. Discussion (0.5pp)

- The general lesson for this venue: a quantity can look basis-free (an SVD, a pseudo-inverse, a projection onto a tangent space) while silently encoding a metric on parameter space that the theory never fixes. This failure mode should be expected wherever a generator is projected onto an overparameterised, rank-deficient tangent space — i.e. essentially always.
- Limitations: two systems, one architecture family, few seeds; the causal effect sizes are noisy (see open items); two-body covers rotation only.
- Future work, explicitly deferred: a subspace-valued attribution reporting an equivalence class rather than a point; disentangling symmetry-specific from general sensitivity beyond the matched-null approach; whether SGD's implicit bias, not merely convergence, drives §4.5.

Optional §5.5 — a bounded, explicitly-scoped demonstration

(Include only if space permits; one figure, one paragraph, placed after the negative results.) In one trained instance at one fixed gauge, an explicit 20-parameter subset (of 4514) shows a causally-verified positive

effect on the symmetry defect beyond a sensitivity-matched control. Framed as an existence proof — such subsets exist and can be verified — with the identity explicitly disclaimed as non-invariant under §3.3–3.5. **Requires** the second-seed replication showing a *different* index set with the same causal signature; without that pairing it reads as the very claim §3–4 refute.

Claims we make / do not make

We claim	We do not claim
`|c_i|`'s concentration is reproduced by nulls and by conditioning	that `c_i` is uninformative about the network in every respect
the parameter-level *identity* is gauge-dependent	that no parameter subset has causal effect
alignment-selected subsets causally affect equivariance	that they are unique, or reproducible across seeds
symmetry-specific concentration falls as equivariance improves	that the symmetry itself weakens (it strengthens by 3 orders)
`E_i`, rank, `P_T g` are invariant	that they answer “which parameters”

Figures (5)

1. The factorisation: `log|c_i|` vs `log□ S_i□` , with `|a_i|` and `|ã_i|` — slopes −0.99 / +1.01 / −0.000.
2. Lorenz curves: true generator vs matched null vs `1/□ S_i□ ²` alone.
3. Cross-gauge overlap matrix (8 gauges), with the disjoint pair highlighted.
4. Causal: matched-control excess by score, with the published score at zero.
5. **Headline** — delocalisation: concentration z and (symm,task) against equivariance error across training.

Claim → evidence → script

Claim	Evidence	Script
conditioning factorisation is exact	residual $6\times10^{\ 1}$	`conditioning_decomposition.py`
null calibration cancels it exactly	residual $4\times10^{\ 1}$	`causal_symmetry_control.py`
gauge dependence	solution-set dim, cross-gauge Jaccard	`gauge_dependence.py`
covariance breaks, 3 sources	departure table	`reparametrisation_covariance.py`
`E_i` invariance	$2.8\times10^{\ 1}$	`reparametrisation_covariance.py`
identifiability vs null floor	Jaccard table, §4.3b	`attribution_stability.py`
rank stable under grid refinement	96→1792 rows, rank 22	`probe_grid_sweep.py`
causal excess, matched control	Δlog D excess by score	`causal_matched_ablation.py`
controlling set non-unique	disjoint-block excess	`disjoint_sets_control.py`
influence diffuse; sufficiency worse than null	n_eff; OMP k*( ) vs nulls	`symmetry_concentration.py`, `symmetry_subspace_sparsity.py`
delocalisation over training	z, (symm,task) vs equivariance	`delocalisation_sweep.py`
Prop. 0 (,g =0)	cos 10 , both systems	`pi_delta_split.py`, `pi_delta_split_two_body.py`

Proofs and verification residuals: `NOTES_gauge_and_conditioning.md`.

Open items before submission

1. **Fig. 5 needs the full checkpoint sweep** — currently two endpoints; the 13-checkpoint run is in progress.
2. **Error bars on the causal excess** — current SEs cover matched-control variation only; the same top-20 gave +0.58 and +0.14 on two runs. Needs variance over perturbation draws and seeds.
3. **Second seed** for §5.5, or drop §5.5.
4. Decide whether two-body goes in the main text or appendix.